

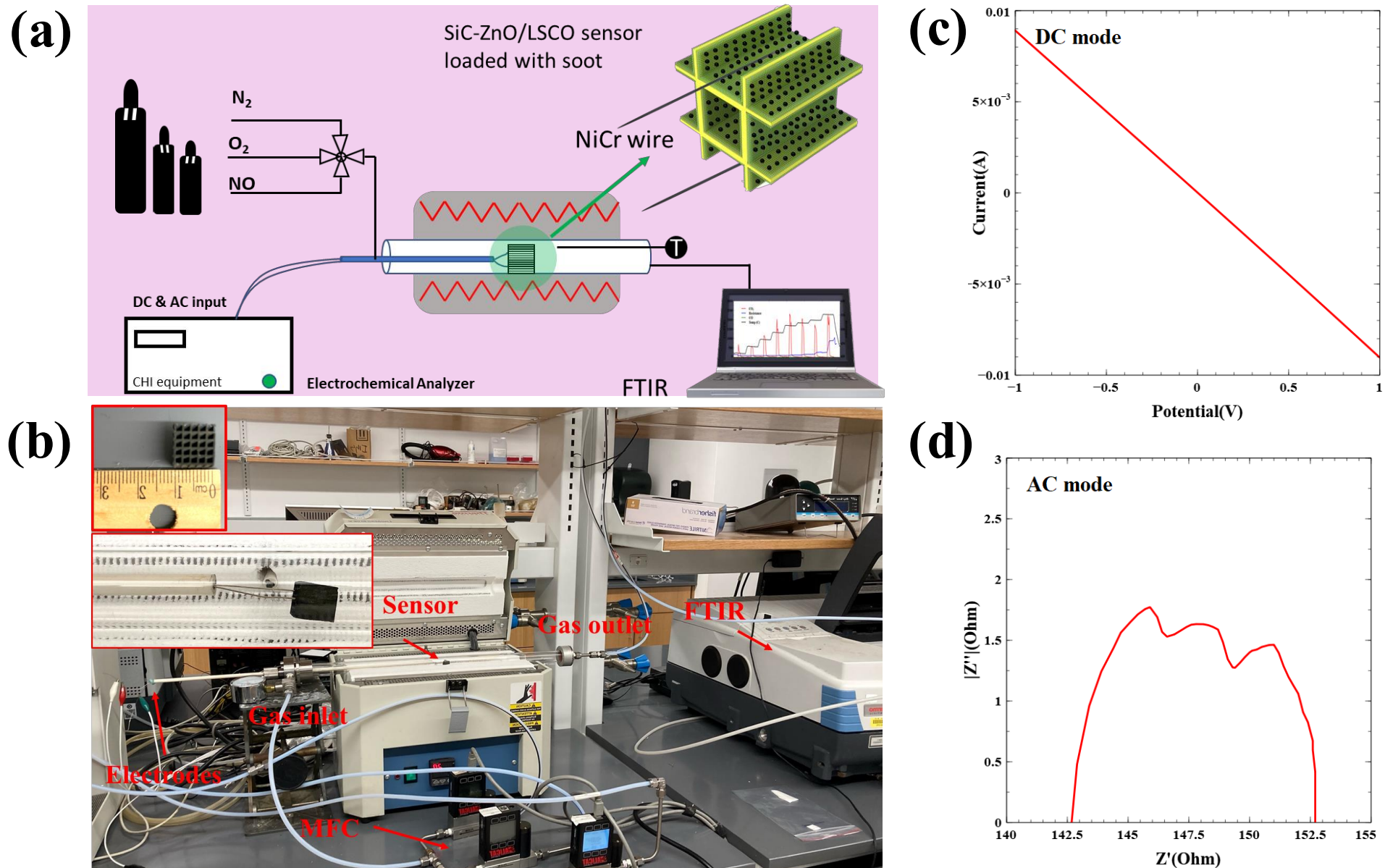
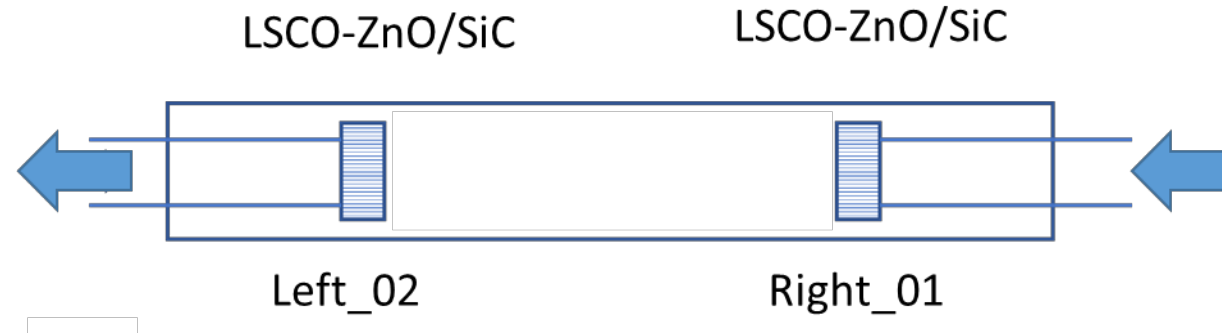


Figure 1. (a) Schematic drawing of the soot sensor testing system and the sensor wiring. (b) Photo of the sensor experimental setup (inset: photo of the SiC-ZnO/LSCO sensor and sensor wiring). (c) and (d) CV curve and Nyquist plot of the SiC-ZnO/LSCO sensor.

Experimental setup and testing parameters-NO_x gas sensing



Temperature: 400, 500C, 600C, Air as purging gas, 5 min of analyte flowing, 10 min Air purging, **9 points in total tested**

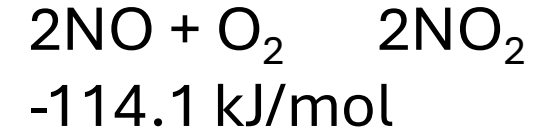
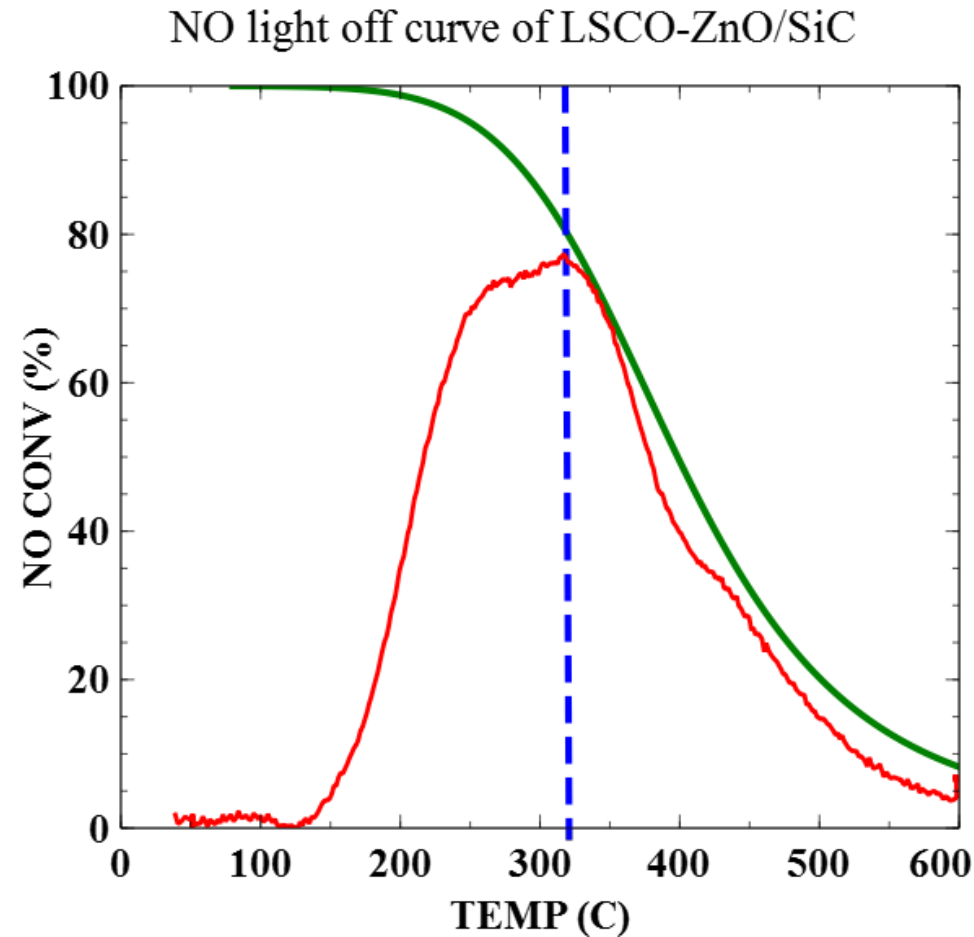
Feed gas condition-1

NO original	NO FLOW	Air flow	TotFlow	NO set ppm
1000	3	497	500	6
	5	495	500	10
	10	490	500	20
	25	475	500	50
	50	450	500	100
	75	425	500	150
	100	400	500	200
	125	375	500	250
	150	350	500	300

Feed gas condition-2

NO ₂ original	NO ₂ FLOW	Air flow	TotFlow	NO ₂ set ppm
1000	3	497	500	6
	5	495	500	10
	10	490	500	20
	20	480	500	40
	40	460	500	80
	75	425	500	150
	100	400	500	200
	125	375	500	250
	150	350	500	300

NO light off curve based on the sensor setup

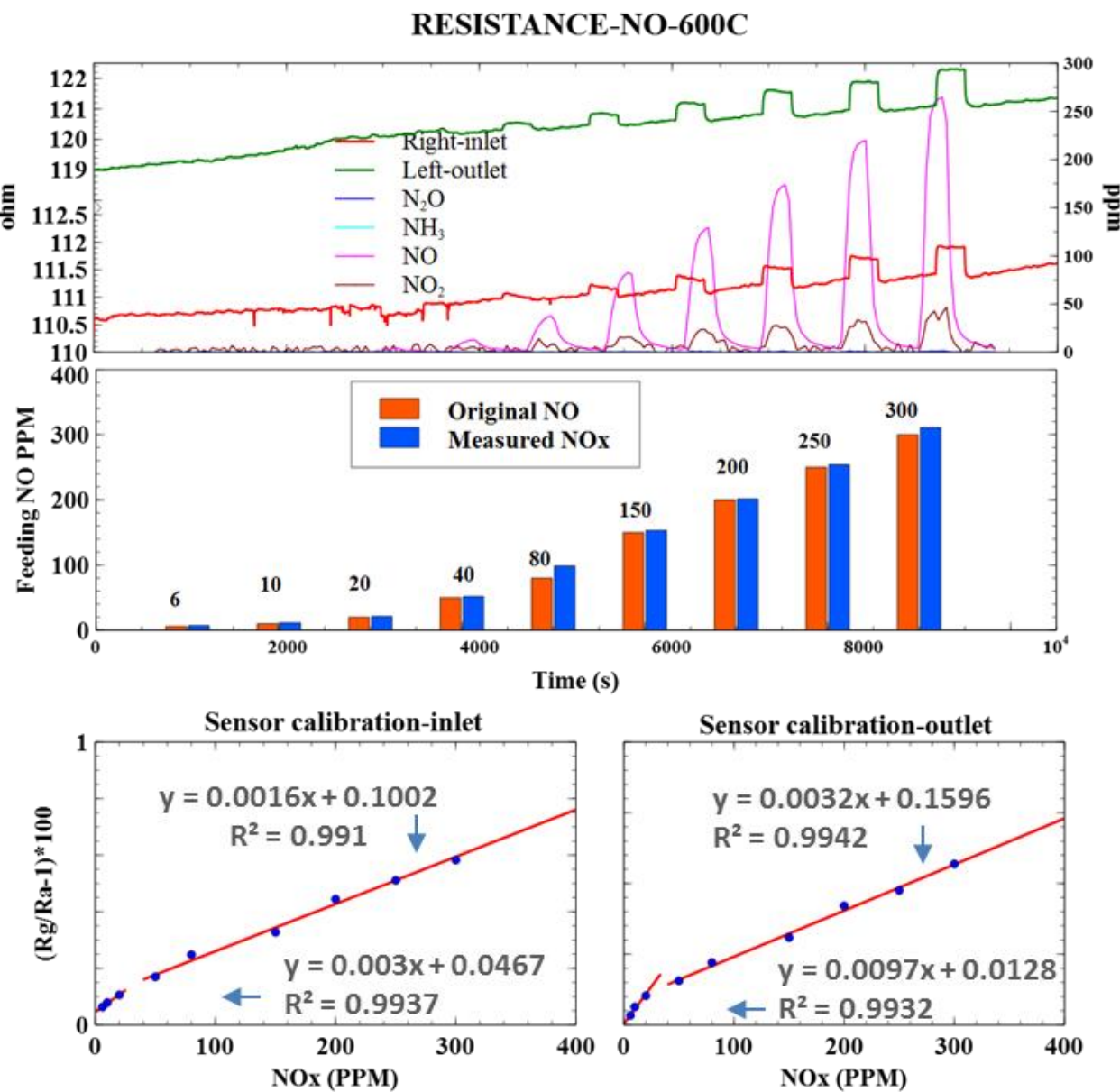


The gas environment is a mixture of NO and NO₂ for sensor

Flow rate: NO+N₂ = 500 sccm, 10% O₂

[NO]=100ppm, T: 50-600 C, ramping rate: 2C/min

DC mode NOx sensing at 600 °C



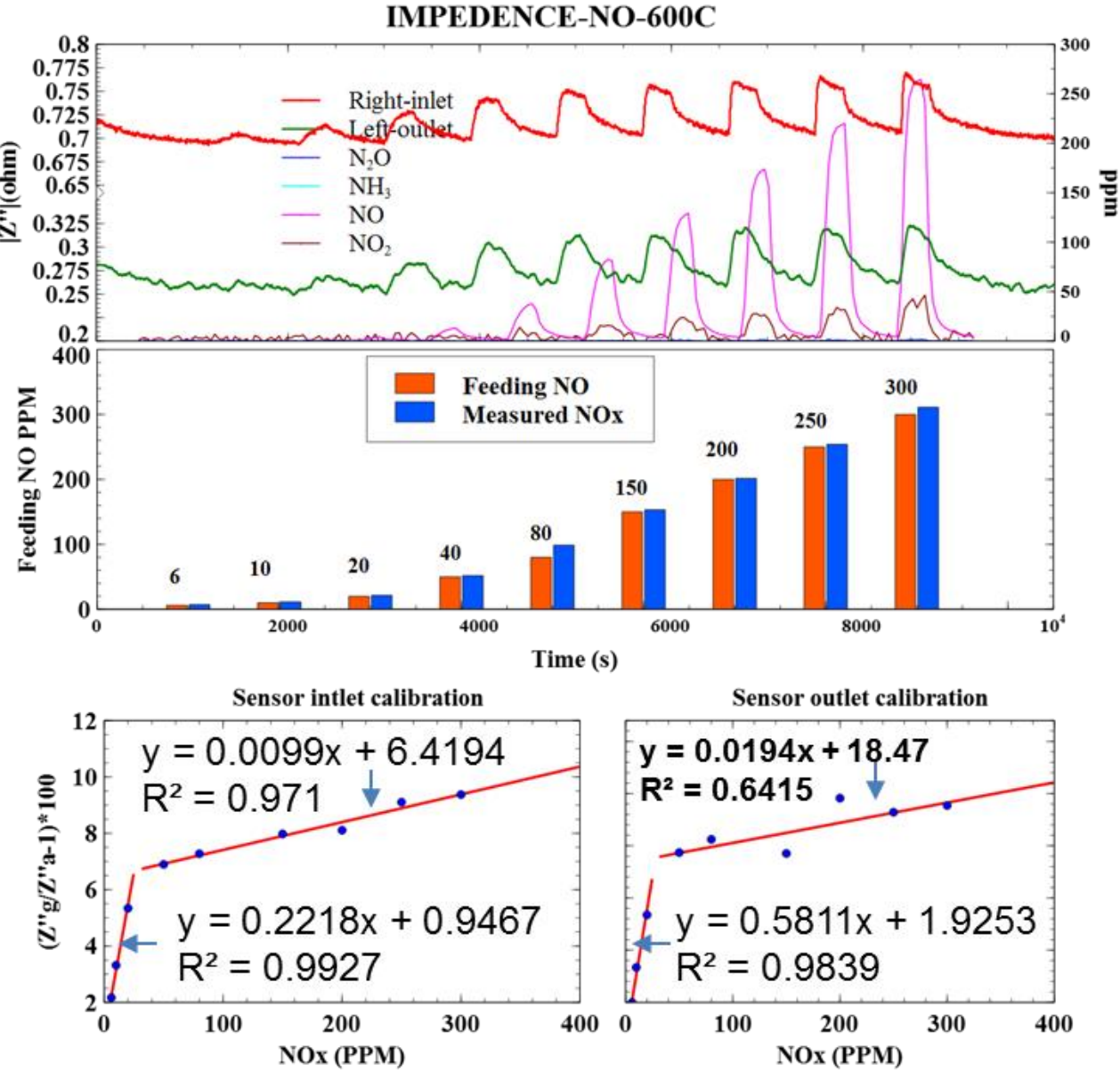
Calibration of NOx with NO and Air feeding in resistance mode. Dynamic and linear calibration of 6-300 ppm of NOx. DC amplitude: 1 V

- 1. Total NOx (NO + NO₂) can be monitored.
- 2. LSCO sensor shows two linear responses for NO_x in the range of 40-300 ppm and 6-20ppm.

$$R_{total} = R_{NO} + R_{NO2} + R_{reaction\ electron\ transfer}$$

How machine learning can be applied to deconvolute different R?

AC mode NOx calibration-600C



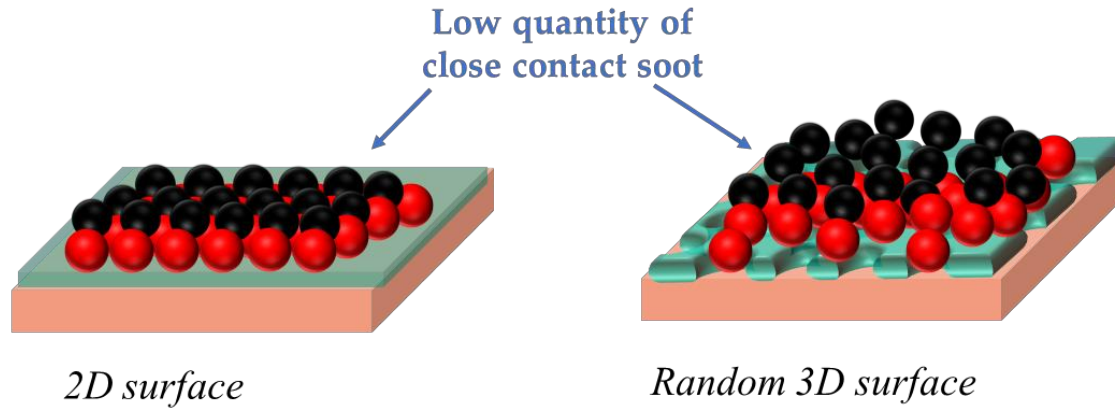
Calibration of NOx with NO and Air feeding in impedance mode. Dynamic and linear calibration of 6-300 ppm of NOx. AC amplitude: 0.2 V, scanning freq: 200 Hz

AC mode shows different calibration behavior that is related to NO, NO_2 concentration and other parameters.

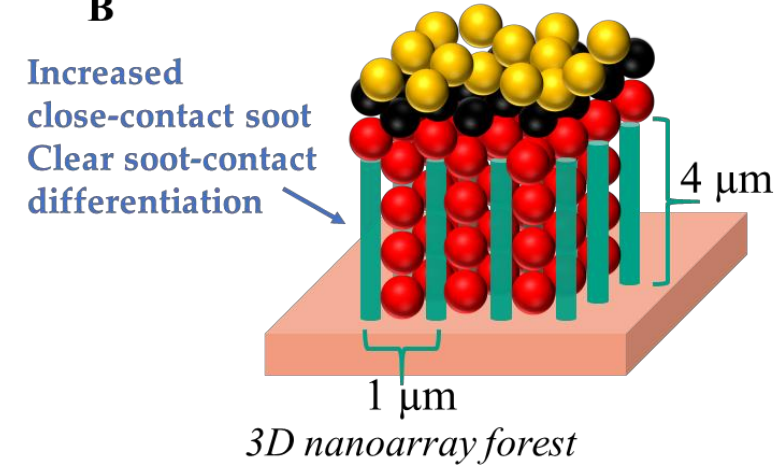
Extra dimension can be used for data deconvolution.

Soot oxidation sensing and monitoring-Solid sensing

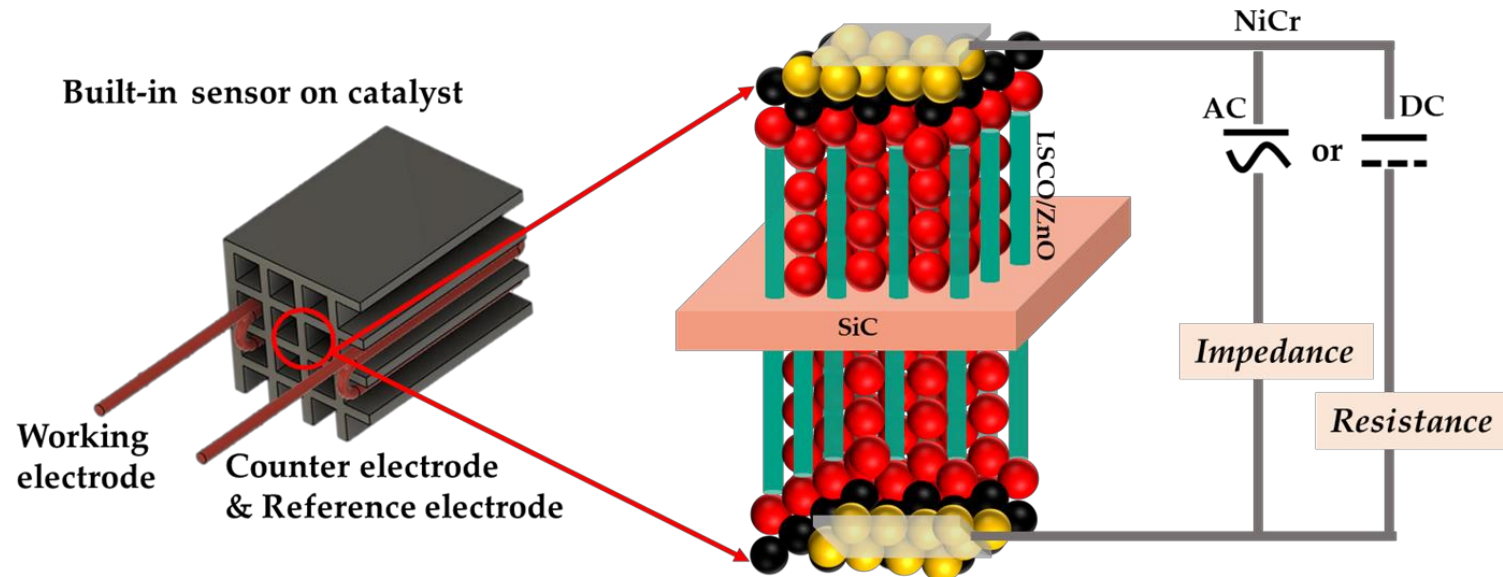
A



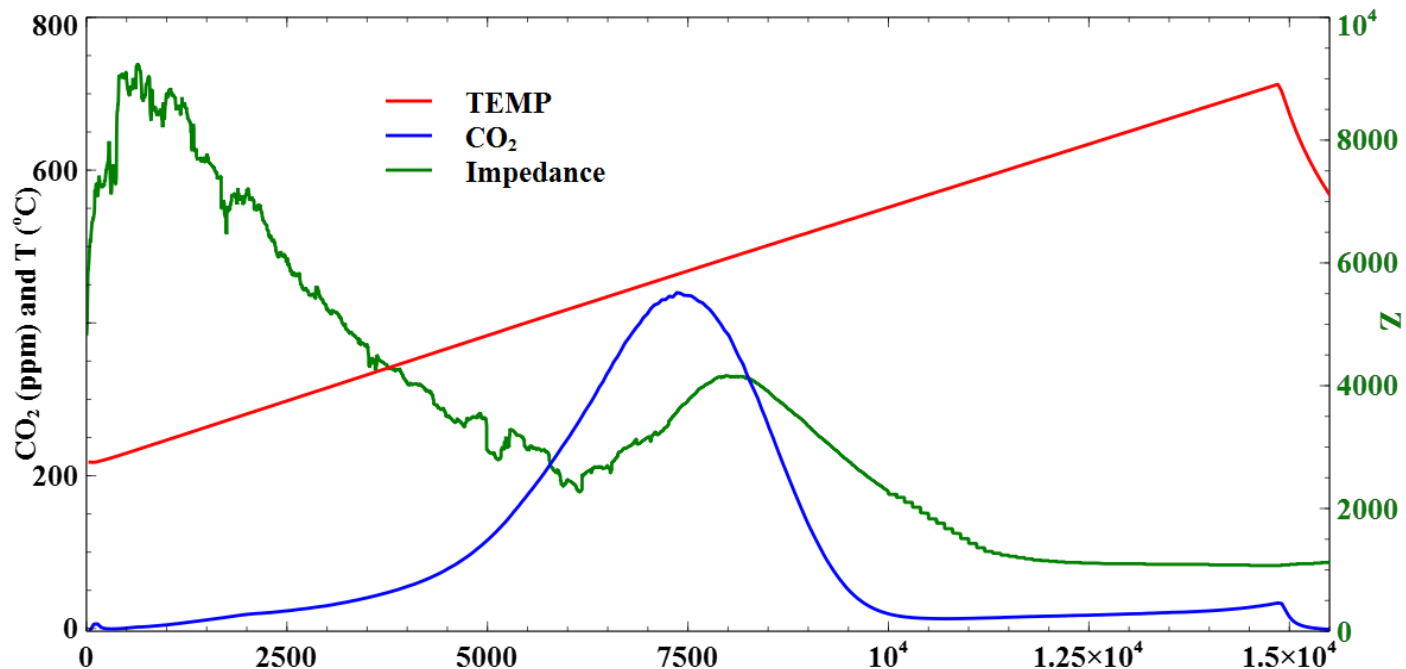
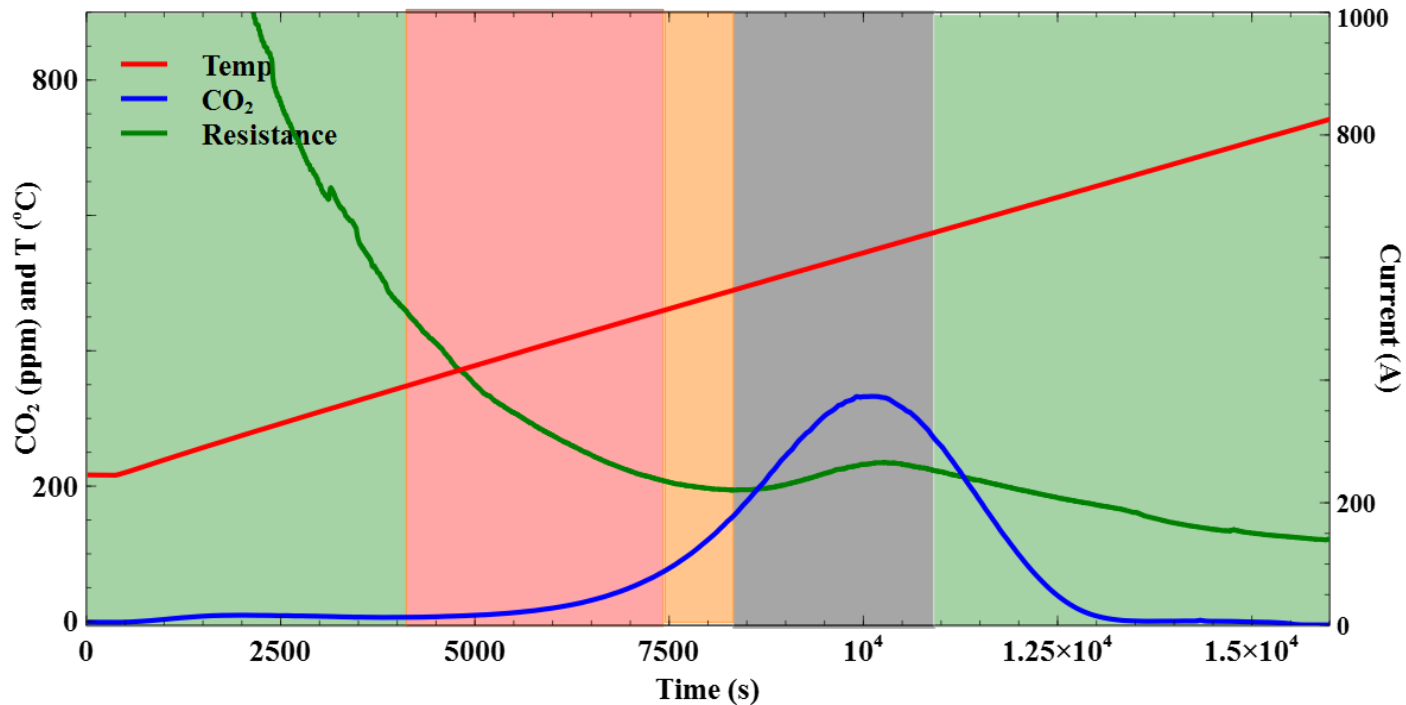
B



C



● Loose contact soot ● Close contact soot ● Surface loose contact soot ■ Supported catalyst ■ SiC substrate



DC mode

Temperature: 200-800 C, 2 C/min
8% of O₂ in N₂

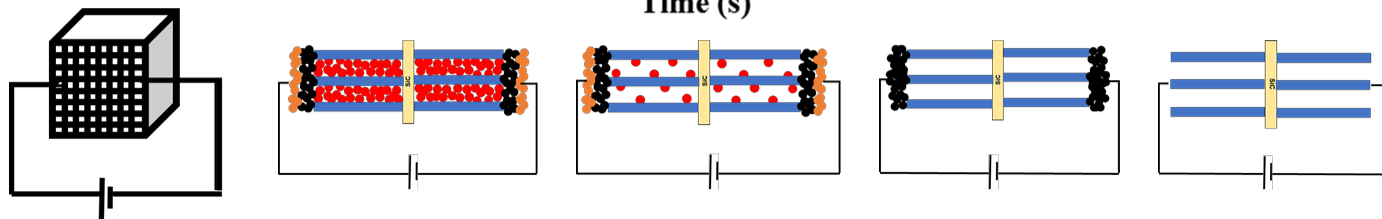
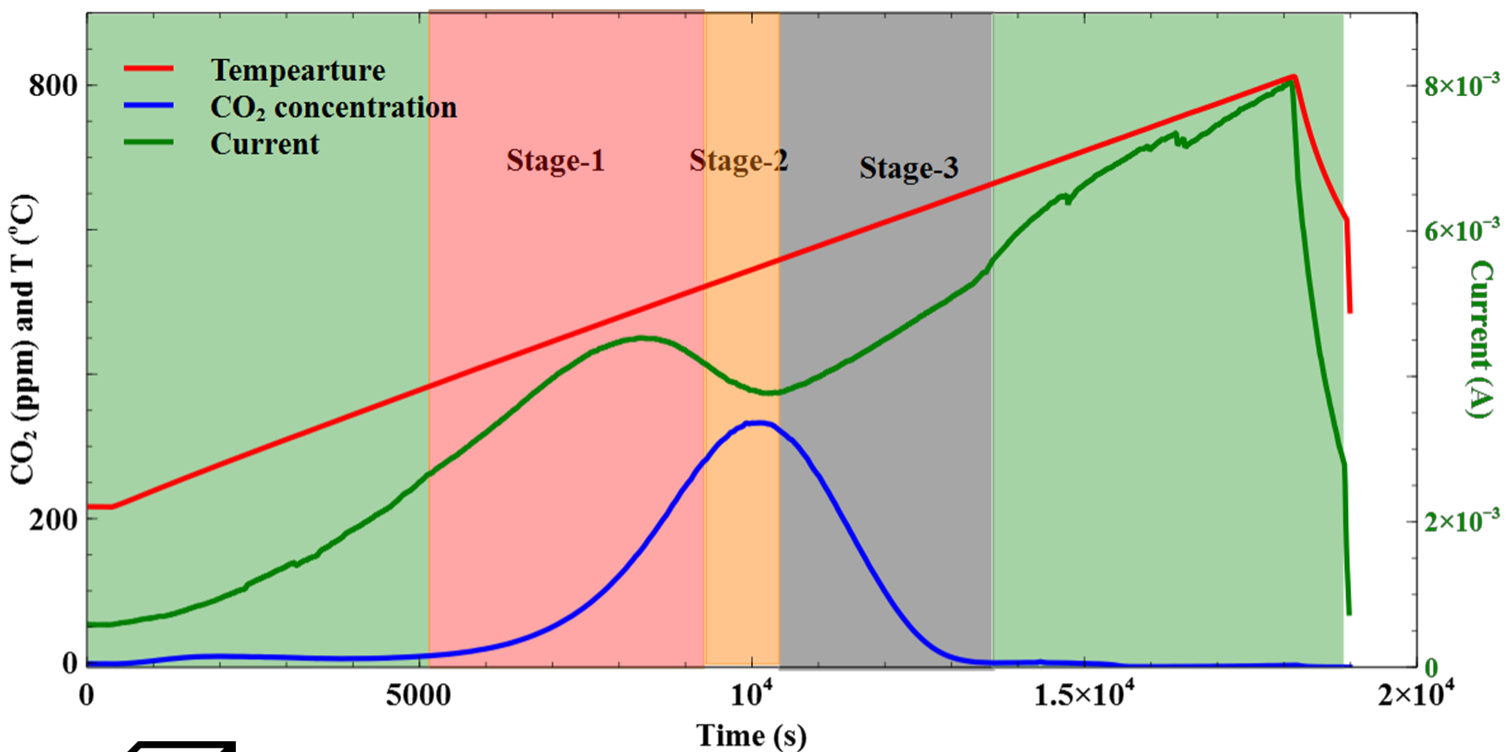
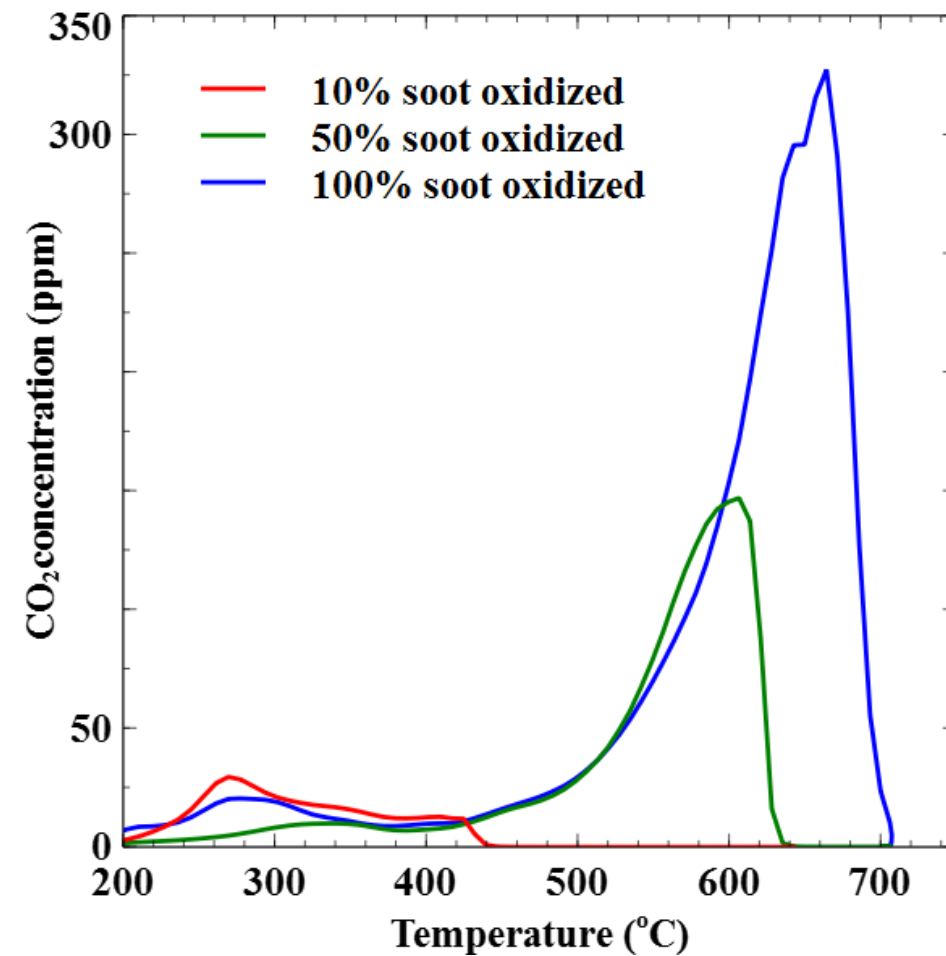
The resistance change is a result of mixed factors, including temperature, soot configuration change and soot reaction rate change and more.

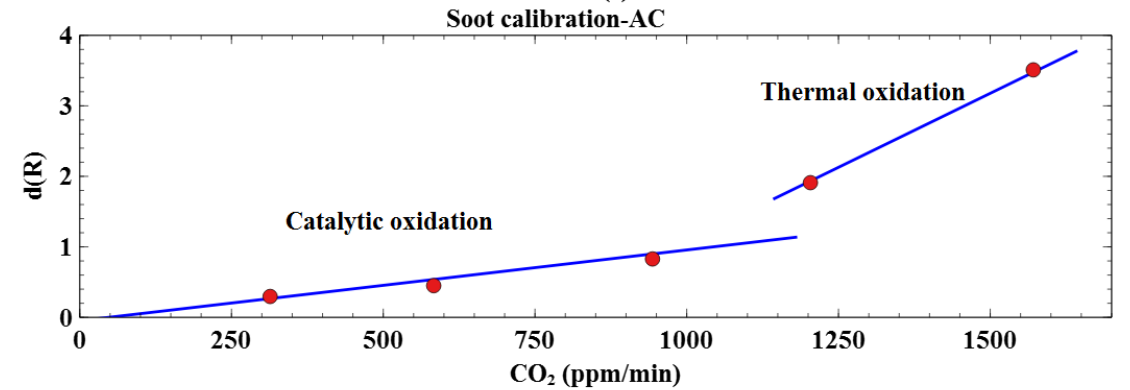
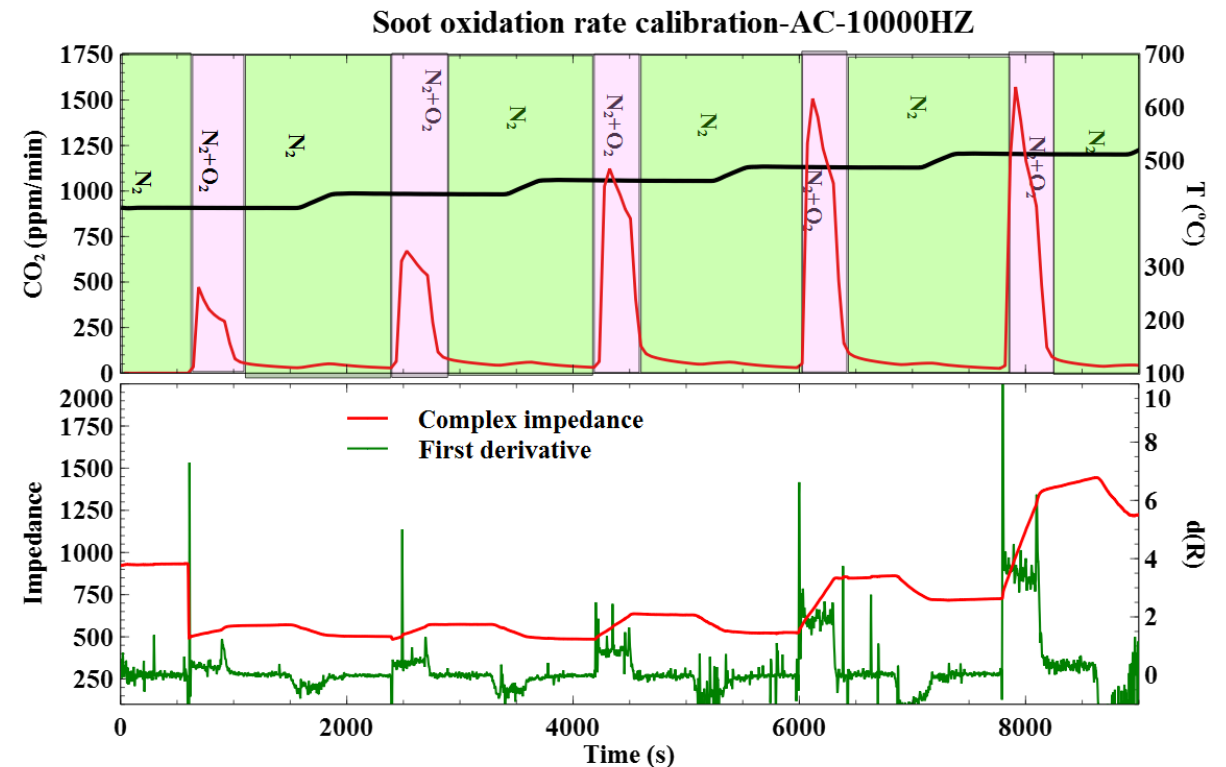
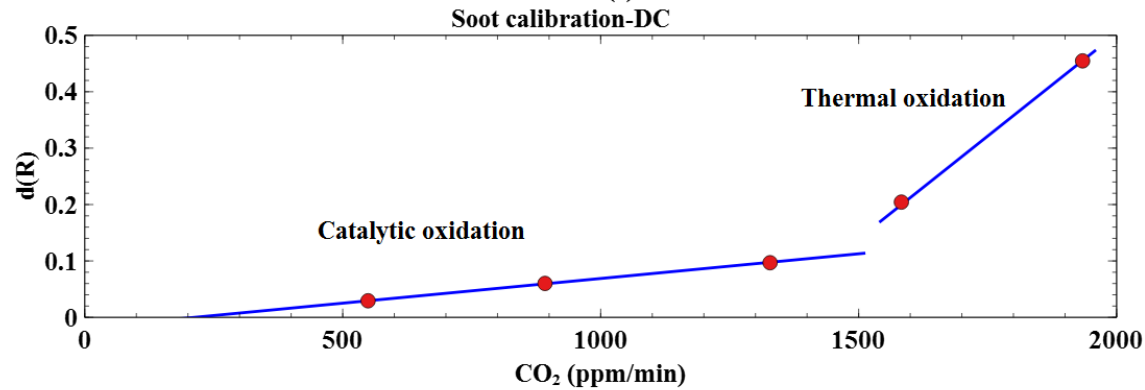
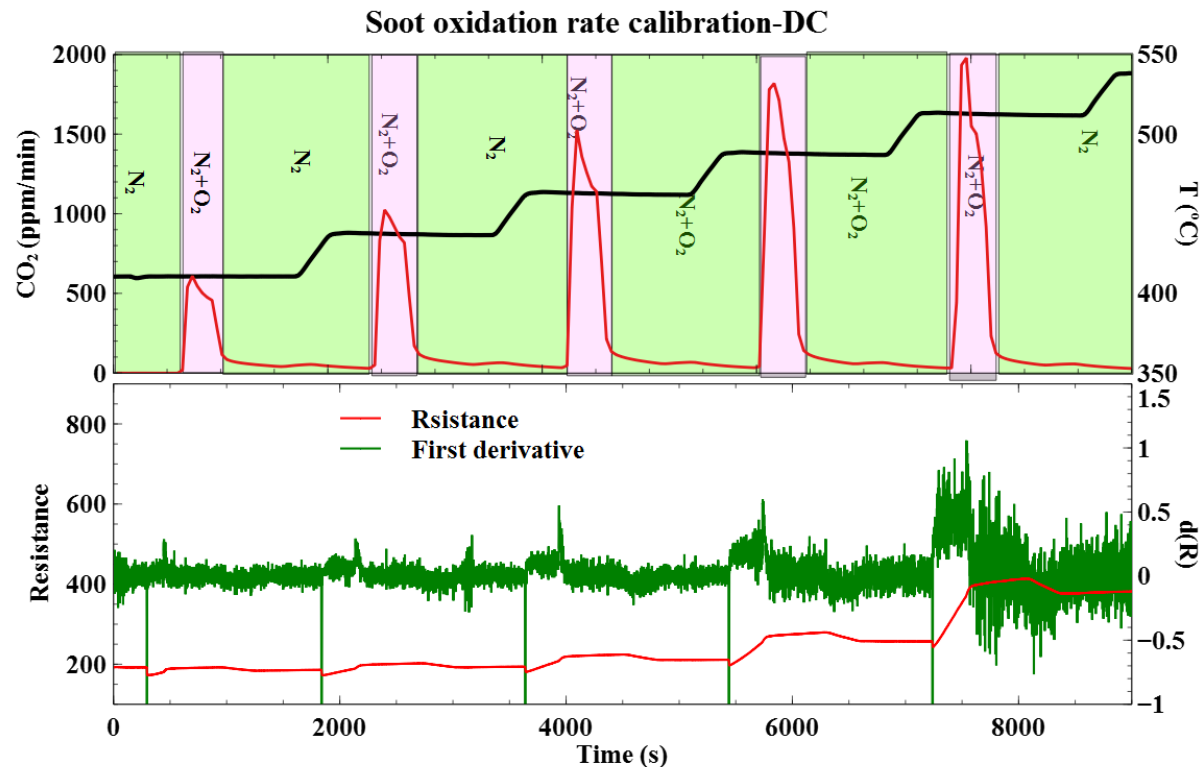
$$R_{\text{total}} = R_T + R_{\text{soot oxidation rate}} + R_{\text{configuration change}} + R_{\text{gas composition}} + R_{\text{else}}$$

$R_{\text{configuration change}}$ make binary assumption for 3 different modes

AC mode

Using alternating current shows the similar trend for the resistance change. How can AI help to deconvolute contribution from different factors

(a)**(b)**

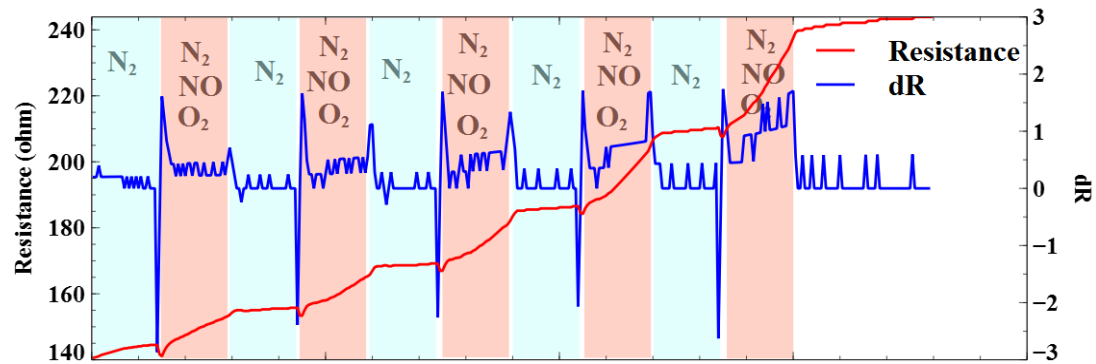


Soot oxidation is conducted at isotherm conditions, while temperature contributes the oxidation rate change

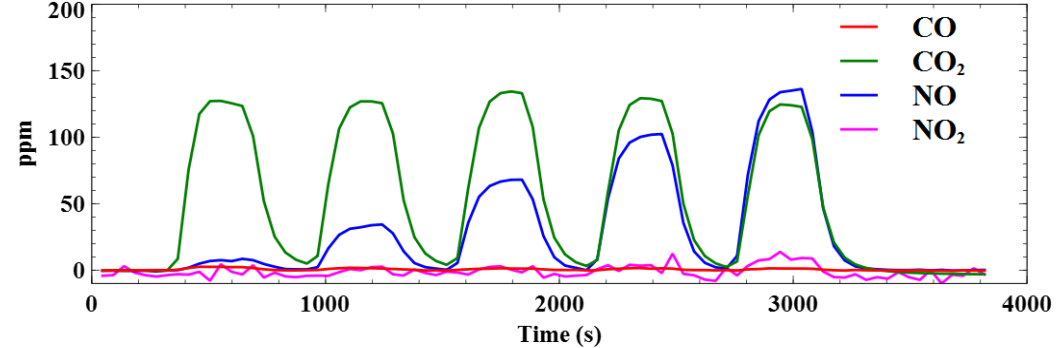
Soot oxidation sensing with DC and AC mode with linear calibration of soot oxidation rate.

Soot oxidation rate and NOx correlation

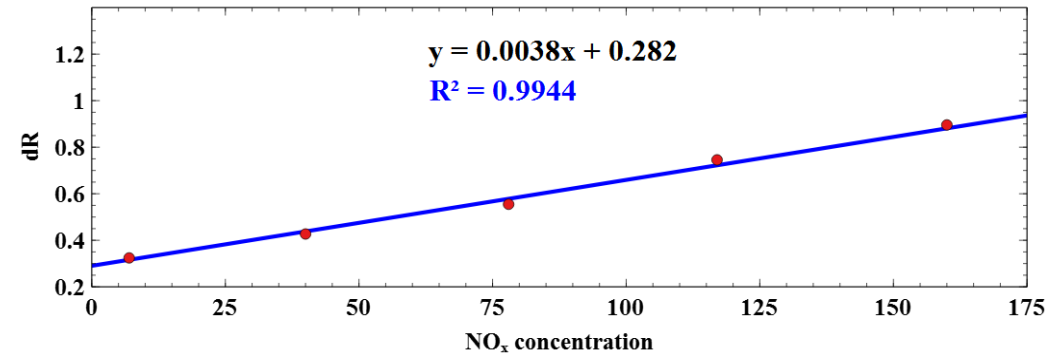
(a)



(b)



(c)

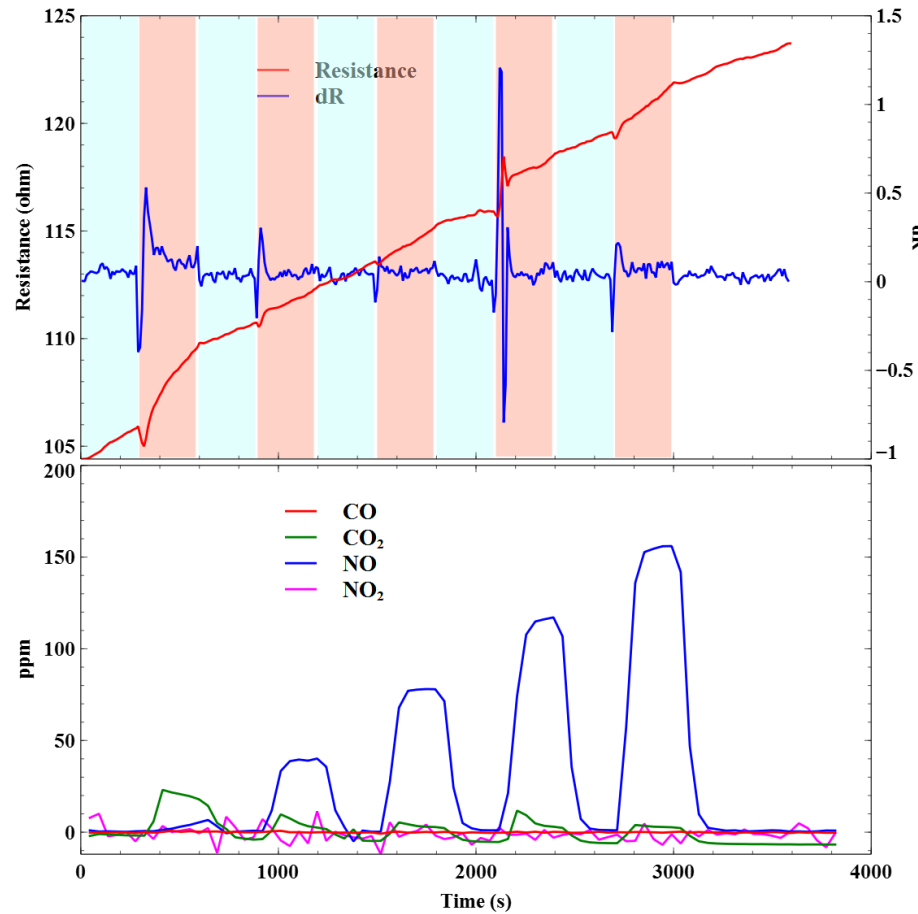


1. Soot oxidation rate shows linear correlation with NOx concentration.

Reaction condition: 450 C, soot is preloaded. NO is pulse loaded for 5 min. NO:10-200 ppm. DC mode.

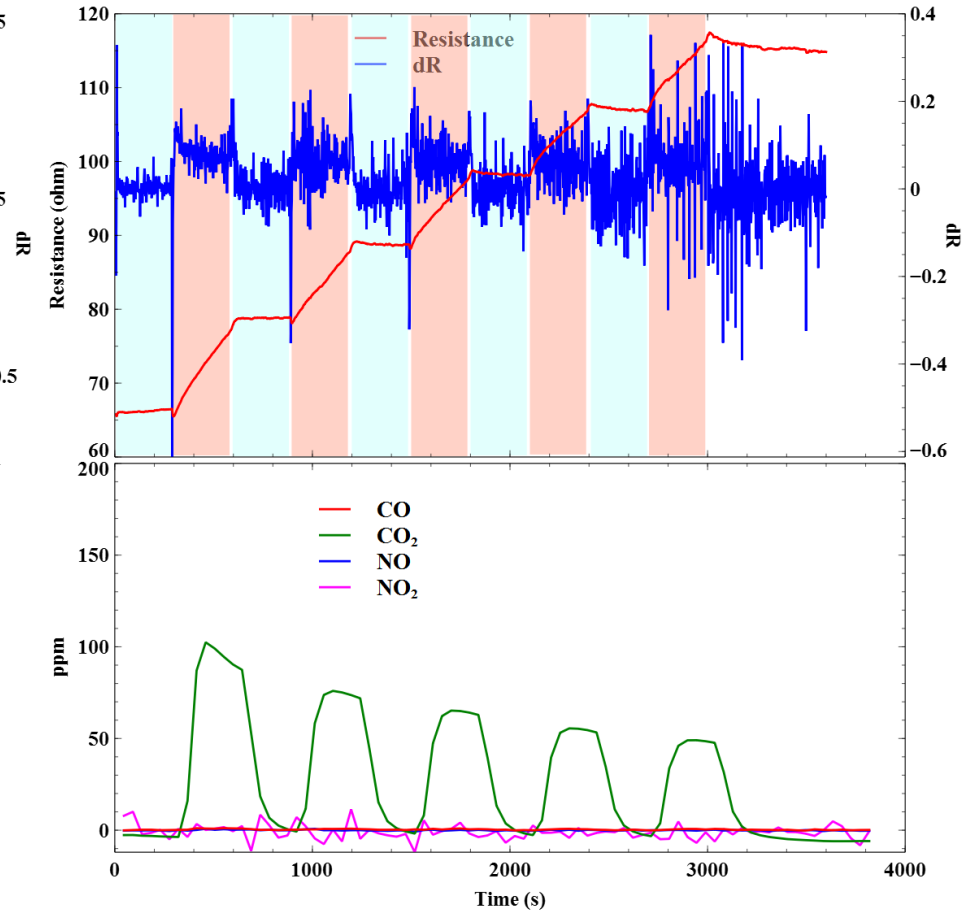
Soot oxidation rate NO and O2 only

NO only



Reaction condition: 450 C, soot is preloaded. NO is pulse loaded for 5 min. NO:10-200 ppm. DC mode.

O2 only



Reaction condition: 450 C, soot is preloaded. 8% O₂ is pulse flowed for 5 min. DC mode.

Summary
O₂ and NO only shows no effect on the soot oxidation rate.